

Force

In physics, a **force** is a push or pull acting upon an object as a result of the object's interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector Quantity** – Forces have both magnitude (strength) and direction.
- **SI Unit** – The standard metric unit for force is the **Newton (N)**.

One Newton is the force required to accelerate a 1 kg mass at a rate of **1 m s^{-2}** .

- **Formula** – Newton's Second Law of Motion:

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$$F = m \cdot a$$

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where F is force, m is mass, and a is acceleration.

Common Types of Forces

Forces are generally grouped into two main categories:

1. Contact Forces

Applied through physical contact between objects. Examples include:

- **Friction** – resistance to relative motion.
- **Tension** – pulling force transmitted through a string or rope.
- **Air resistance** – drag force opposing motion through a fluid.

- **Normal force** – perpendicular reaction force from a surface.
- **Applied push/pull** – direct human or mechanical forces.

2. Non-Contact (or Long-Range) Forces

Act over a distance without physical contact. Examples include:

- **Gravitational force** – attraction between masses.
- **Electrostatic force** – attraction/repulsion between charged particles.
- **Magnetic force** – interaction between magnetic fields and currents.
- **Nuclear forces** – strong and weak forces binding atomic nuclei.

> **Note:** The above definitions and formulas are standard in introductory physics and are widely used in textbooks and academic references.